

Financial Statistics - Coursework

Group 30

2025-11-23

Table of contents

1	Question 1 - CAPM	3
1.1	Prepare the Dataset	3
1.2	Estimate two linear regression models	4
1.3	F-test for $H_0 : \beta_1 = \beta_2$	5
1.4	t-test for $H_0 : \alpha = 0$	5
2	Question 2 - probability of a positive asset return	5
2.1	Prepare the Dataset	5
2.2	Estimate the linear model	6
2.3	Compute Z and Find best predictors	6
2.4	Search optimal that maximizes Z	9
3	Question 3 - CIR model for the term structure of interest rate	10
3.1	Prepare the Dataset	10

Table of contents

List of Figures

1	Z(alpha) over alpha	10
2	1-Month Interest Rates: US vs UK	11
3	US 1-Month Interest Rate: Actual vs Predicted	19
4	UK 1-Month Interest Rate: Actual vs Predicted	20

List of Tables

1	Preview of the Q1 dataset	3
2	F-test Outcome	5
3	Preview of the Q2 dataset	6
4	Binary Response Regression Summary	7
5	Best Binary Response Regression Model Results	8
6	Preview of the Q3 dataset	10
7	US Interest Rate Model Estimation Results	14
8	UK Interest Rate Model Estimation Results	15
9	Model Comparison Results	16
10	Asymptotic Covariance Matrices	17

```
# Load required libraries
suppressPackageStartupMessages({
  library(readxl)
  library(lubridate)
  library(readr)
  library(utils)
  library(quantmod)
  library(tidyverse)
  library(numDeriv)
  library(MASS)
  library(ggplot2)
})
```

1 Question 1 - CAPM

1.1 Prepare the Dataset

```
# Use the dataset data coursework Q1
data_q1 <- read_excel("data_coursework_Q1.xls")

data_q1$date <- make_date(year = data_q1$year_, month = data_q1$month_, day = 1)
data_q1$IBM <- parse_number(data_q1$IBM)
df_q1 <- data_q1[, c("date", "SP500", "IBM", "1-month Tbill")]

colnames(df_q1)[colnames(df_q1) == "1-month Tbill"] <- "Rf"
colnames(df_q1)[colnames(df_q1) == "IBM"] <- "Ri"

df_q1$Rm <- (df_q1$SP500 / dplyr::lag(df_q1$SP500)) - 1
df_q1$Rf <- df_q1$Rf / 12

head(df_q1)
```

A tibble: 6 × 5

Table 1: Preview of the Q1 dataset

date <date>	SP500 <dbl>	Ri <dbl>	Rf <dbl>	Rm <dbl>
1960-01-01	58.03	NA	0.02750000	NA
1960-02-01	55.78	NA	0.02416667	-0.038773048
1960-03-01	55.02	NA	0.02916667	-0.013624955
1960-04-01	55.73	NA	0.01583333	0.012904398
1960-05-01	55.22	NA	0.02250000	-0.009151265
1960-06-01	57.26	NA	0.02000000	0.036943137

```
# Construct the measures

# excess return for IBM (Ri - Rf)
df_q1$ri_rf <- df_q1$Ri - df_q1$Rf
# excess market return (Rm - Rf)
df_q1$rm_rf <- df_q1$Rm - df_q1$Rf
# squared market excess return
df_q1$rm_rf_sq <- df_q1$rm_rf^2

# D_t = 1 if market excess return > 0, 0 otherwise
df_q1$D <- ifelse(df_q1$rm_rf > 0, 1, 0)
df_q1$D_rm <- df_q1$D * df_q1$rm_rf
df_q1$Dinv_rm <- (1 - df_q1$D) * df_q1$rm_rf
```

1.2 Estimate two linear regression models

```
# Model 1: Standard CAPM
# ri_rf = alpha + beta * rm_rf + u
capm_model <- lm(ri_rf ~ rm_rf, data = df_q1)

# show summary
summary(capm_model)

# Model 2: Extended CAPM
# ri_rf = alpha + beta1*D_rm + beta2*Dinv_rm + beta3*rm_rf_sq + u
ext_model <- lm(ri_rf ~ D_rm + Dinv_rm + rm_rf_sq, data = df_q1)

# show summary
summary(ext_model)
```

Call:

```
lm(formula = ri_rf ~ rm_rf, data = df_q1)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.878	-4.733	-2.628	7.244	19.250

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.8675	0.5256	18.775	<2e-16 ***
rm_rf	-5.0450	9.0589	-0.557	0.578

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.112 on 346 degrees of freedom

(24 observations deleted due to missingness)

Multiple R-squared: 0.0008956, Adjusted R-squared: -0.001992

F-statistic: 0.3101 on 1 and 346 DF, p-value: 0.578

Call:

```
lm(formula = ri_rf ~ D_rm + Dinv_rm + rm_rf_sq, data = df_q1)
```

Residuals:

Min	1Q	Median	3Q	Max
-9.040	-4.880	-2.794	6.682	19.697

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	9.0742	0.8315	10.913	<2e-16 ***
D_rm	18.0443	51.6197	0.350	0.7269
Dinv_rm	-49.9093	28.6993	-1.739	0.0829 .
rm_rf_sq	-350.2208	196.2948	-1.784	0.0753 .

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.097 on 344 degrees of freedom

(24 observations deleted due to missingness)

Multiple R-squared: 0.01081, Adjusted R-squared: 0.002179

F-statistic: 1.253 on 3 and 344 DF, p-value: 0.2906

1.3 F-test for $H_0 : \beta_1 = \beta_2$

```
# Restricted model: force beta1 = beta2
restricted_model <- lm(ri_rf ~ rm_rf + rm_rf_sq, data = df_q1)

# F-test between restricted and unrestricted models
anova(restricted_model, ext_model)
```

A anova: 2×6

Table 2: F-test Outcome

	Res.Df <dbl>	RSS <dbl>	Df <dbl>	Sum of Sq <dbl>	F <dbl>	Pr(>F) <dbl>
1	345	17373.33	NA	NA	NA	NA
2	344	17328.28	1	45.04091	0.8941493	0.3450191

The F-test compares the restricted symmetric CAPM model with the unrestricted asymmetric beta specification. The test statistic is $F = 0.894$ with a corresponding **p-value = 0.345**. Because the p-value is well above conventional significance levels (1%, 5%, or 10%), we fail to reject the null hypothesis that $\beta_1 = \beta_2$.

This implies that the market beta of IBM does not differ significantly between up-market periods ($R_m - R_f > 0$) and down-market periods ($R_m - R_f \leq 0$). In other words, the data do not provide evidence of return asymmetry or different systematic risk exposures across market states. The standard CAPM with a single beta coefficient appears sufficient for this sample.

1.4 t-test for $H_0 : \alpha = 0$

```
# Extract summary table
capm_summary <- summary(capm_model)$coefficients

# Alpha estimate
alpha_est <- capm_summary["(Intercept)", "Estimate"]
alpha_t <- capm_summary["(Intercept)", "t value"]
alpha_p <- capm_summary["(Intercept)", "Pr(>|t|)"]

alpha_est
alpha_t
alpha_p
```

9.86746894909065

18.7745349820389

1.00130758429436e-54

The CAPM regression shows that the estimated alpha (intercept) for IBM is **9.87**, with a **t-statistic of 18.77** and a p-value effectively 0. This indicates that the abnormal return (Jensen's alpha) is highly significant, meaning IBM delivered a large positive excess return that cannot be explained by market movements alone during the sample period. The null hypothesis $\alpha = 0$ is strongly rejected.

2 Question 2 - probability of a positive asset return

2.1 Prepare the Dataset

```
# Load data
data <- read_excel("data_coursework_Q2.xls", sheet = 1)
data <- na.omit(data)
head(data)
```

```
# Separate response variable and regressors
returns <- data$ESR
regressors <- data[, -which(names(data) == "ESR" | names(data) == "mth")]
```

A tibble: 6 × 12

Table 3: Preview of the Q2 dataset

mth <chr>	ESR <dbl>	USR <dbl>	EIF <dbl>	ERECB <dbl>	EFX <dbl>	EDY <dbl>	ESP <dbl>	UIF <dbl>	URR <dbl>	UDY <dbl>	UOIL <dbl>
31- gen-85	0.0559	0.0690	0.0209	1.6094	0.0025	1.1663	2.206	0.0396	2.1668	1.4951	0.0210
31106	0.0331	0.0066	0.0235	1.6582	0.0556	1.1474	2.374	0.0347	2.1679	1.4929	0.0094
31137	-	-	0.0249	1.6582	-	1.1474	2.142	0.0354	2.1494	1.5041	0.0582
	0.0027	0.0089			0.0813						
31167	0.0226	-	0.0250	1.6582	0.0049	1.1569	2.023	0.0372	2.1494	1.5173	-
		0.0057									0.0269
31- mag- 85	0.0424	0.0504	0.0236	1.6582	-	1.1282	1.856	0.0352	2.0334	1.4679	0.0108
					0.0127						
30- giu-85	0.0263	0.0116	0.0193	1.6582	-	1.1019	1.753	0.0351	2.0732	1.4563	-
					0.0099						0.0347

2.2 Estimate the linear model

```
# Create binary response variable
Y <- ifelse(returns > 0, 1, 0)

# Prepare lead-lag data
Y_lead <- Y[-1]
X_lag <- regressors[-length(Y), ]

reg_data <- data.frame(Y_lead, X_lag)

ols_model <- lm(Y_lead ~ ., data = reg_data)
summary(ols_model)
```

The model is **statistically significant** ($p < 0.001$), and although the adjusted R^2 of around 10% indicates a modest explanatory power, it suggests that the included predictors have a meaningful impact on the likelihood of achieving a positive return. Several predictors, including EIF, EDY, and URR, significantly increase the probability of a positive return, while variables such as ERECB, UIF, and UOIL significantly reduce it.

2.3 Compute Z and Find best predictors

```
# models and search for best predictors combination
# Compute predicted probabilities
P_hat <- predict(ols_model, type = "response")

# Construct Z_t(alpha=1/2)
alpha <- 0.5
returns_t <- returns[-1]
Z_t <- ifelse((P_hat > alpha & returns_t > 0) |
             (P_hat <= alpha & returns_t <= 0), 1, 0)

# overall hit ratio
Z_alpha <- mean(Z_t)
cat("Z(alpha = 0.5) for full model =", Z_alpha, "\n")
```

Table 4: Binary Response Regression Summary

Call:

```
lm(formula = Y_lead ~ ., data = reg_data)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.9505	-0.4431	0.1862	0.3995	0.9720

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.44284	0.30703	1.442	0.150325
USR	1.04566	0.62401	1.676	0.094915 .
EIF	8.59691	3.88055	2.215	0.027541 *
ERECB	-0.62801	0.15006	-4.185	3.83e-05 ***
EFX	0.94970	0.88001	1.079	0.281431
EDY	0.58948	0.28929	2.038	0.042525 *
ESP	-0.06447	0.03875	-1.664	0.097318 .
UIF	-12.32022	3.70983	-3.321	0.001017 **
URR	0.42052	0.09377	4.485	1.07e-05 ***
UDY	0.10158	0.16057	0.633	0.527482
UOIL	-0.95801	0.28501	-3.361	0.000884 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4675 on 279 degrees of freedom

Multiple R-squared: 0.1352, Adjusted R-squared: 0.1042

F-statistic: 4.363 on 10 and 279 DF, p-value: 1.081e-05

```
# Search for best combination of predictors that maximizes Z(1/2)
predictor_names <- names(X_lag)
k <- length(predictor_names)

cat("Number of predictors:", k, "\n")

compute_Z <- function(vars) {
  if (length(vars) == 0) {
    model <- lm(Y_lead ~ 1, data = reg_data)
  } else {
    fmla <- as.formula(paste("Y_lead ~", paste(vars, collapse = "+")))
    model <- lm(fmla, data = reg_data)
  }
  Ph <- predict(model, type = "response")
  Zt <- ifelse((Ph > alpha & returns_t > 0) |
              (Ph <= alpha & returns_t <= 0), 1, 0)
  return(list(Z = mean(Zt), model = model, vars = vars))
}

best <- list(Z = -Inf, vars = NULL, model = NULL)

for (m in 1:k) {
  combs <- combn(predictor_names, m, simplify = FALSE)
  for (vars in combs) {
    res <- compute_Z(vars)
    if (res$Z > best$Z) {
      best <- res
    }
  }
}
}
```

Table 5: Best Binary Response Regression Model Results

```
print(summary(best$model))
```

Call:

```
lm(formula = fmla, data = reg_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.9325	-0.4699	0.2351	0.4077	0.7228

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.29068	0.19988	1.454	0.146976
USR	1.23219	0.63004	1.956	0.051488 .
EIF	5.83448	3.58057	1.629	0.104331
ERECB	-0.53419	0.12939	-4.129	4.82e-05 ***
EFX	1.00195	0.89248	1.123	0.262544
EDY	0.70824	0.18753	3.777	0.000194 ***
ESP	-0.06407	0.03804	-1.684	0.093198 .
UIF	-9.83058	3.69086	-2.663	0.008180 **
URR	0.39804	0.08565	4.647	5.18e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4756 on 281 degrees of freedom

Multiple R-squared: 0.09852, Adjusted R-squared: 0.07285

F-statistic: 3.839 on 8 and 281 DF, p-value: 0.0002641

```
# compare with intercept-only model
res0 <- compute_Z(character(0))
if (res0$Z > best$Z) {
  best <- res0
}
cat("Best Z(alpha=0.5) =", best$Z, "\n")
cat("Best predictor combination:\n")
print(best$vars)
```

Z(alpha = 0.5) for full model = 0.6655172

Number of predictors: 10

Best Z(alpha=0.5) = 0.6896552

Best predictor combination:

[1] "USR" "EIF" "ERECB" "EFX" "EDY" "ESP" "UIF" "URR"

The model selected through maximizing the $Z(\alpha=0.5)$ achieves a performance of **0.6897**, which is higher than the full model's value of 0.6655. This improvement indicates that removing less informative predictors enhances forecasting accuracy. The optimal set of eight predictors—USR, EIF, ERECB, EFX, EDY, ESP, UIF, and URR—captures the majority of the predictive power while avoiding unnecessary noise from weaker variables. Statistically, the reduced model remains jointly significant (F-statistic $p < 0.001$), confirming that the selected predictors provide meaningful explanatory value. Several variables, such as ERECB, EDY, UIF, and URR, exhibit strong statistical significance. Although the adjusted R^2 is 7.3%, slightly lower than the full model.

2.4 Search optimal α that maximizes Z

```
# Extract predictions from the best model found earlier
P_best <- predict(best$model, type = "response")

# Search grid for alpha
alpha_grid <- seq(0, 1, length.out = 2001)

Z_alpha_vec <- numeric(length(alpha_grid))

for (i in seq_along(alpha_grid)) {
  a <- alpha_grid[i]
  Z_t_a <- ifelse((P_best > a & returns_t > 0) |
                 (P_best <= a & returns_t <= 0), 1, 0)
  Z_alpha_vec[i] <- mean(Z_t_a)
}

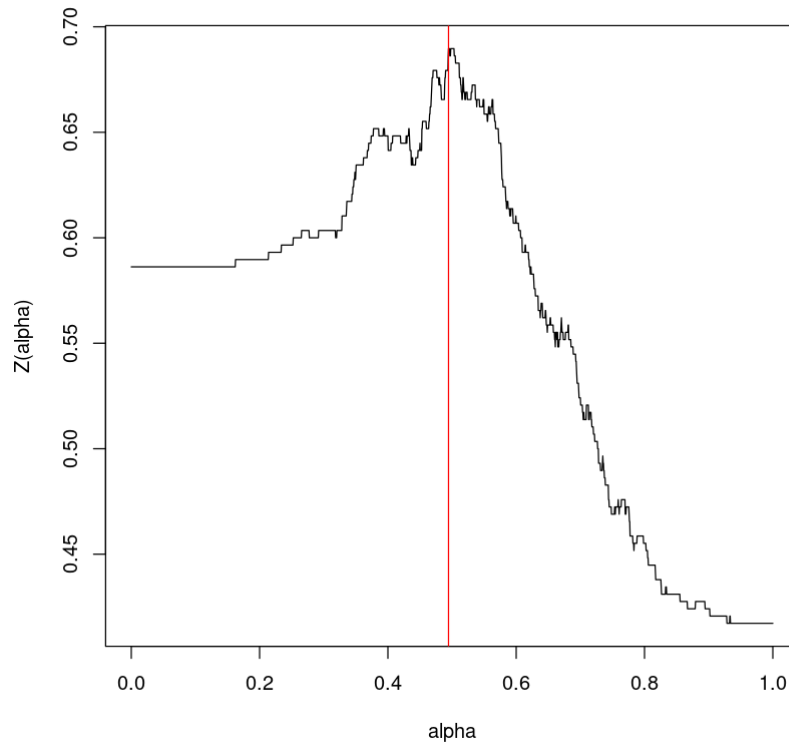
# find maximizing alpha
idx_max <- which.max(Z_alpha_vec)
alpha_best <- alpha_grid[idx_max]
Z_best_alpha <- Z_alpha_vec[idx_max]

cat("Optimal alpha that maximizes Z(alpha):\n")
cat("alpha* =", alpha_best, "\n")
cat("Maximized Z(alpha*) =", Z_best_alpha, "\n")

# Plot Z(alpha)
plot(alpha_grid, Z_alpha_vec, type="l",
      xlab="alpha", ylab="Z(alpha)")
abline(v = alpha_best, col = "red", lwd = )
```

Optimal alpha that maximizes Z(alpha):
alpha* = 0.4945
Maximized Z(alpha*) = 0.6896552

Figure 1: $Z(\alpha)$ over α



3 Question 3 - CIR model for the term structure of interest rate

3.1 Prepare the Dataset

```
# Load the data
data <- read_excel("data_coursework_Q3.xls") %>%
  rename(date = `...1`, r_US = `T r_$`, r_UK = `T r_£`) %>%
  arrange(date) %>%
  drop_na() %>%
  mutate(across(c(r_US, r_UK), ~ ifelse(.x <= 0, NA_real_, .x))) %>%
  drop_na()

head(data)
```

A tibble: 6 × 3

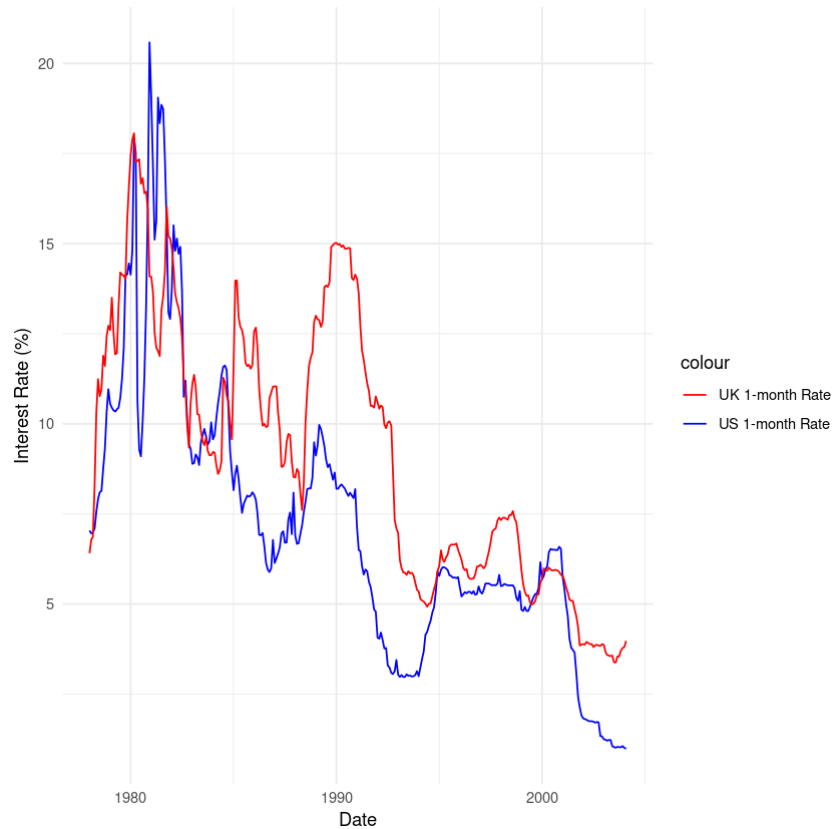
Table 6: Preview of the Q3 dataset

date <dtm>	r_US <dbl>	r_UK <dbl>
1978-01-01	7.04	6.41
1978-02-01	6.96	6.79
1978-03-01	6.99	6.86
1978-04-01	7.09	8.27
1978-05-01	7.55	10.27
1978-06-01	7.91	11.24

```
# Plot interest rates of US and UK
ggplot(data, aes(x = date)) +
  geom_line(aes(y = r_US, color = "US 1-month Rate")) +
```

```
geom_line(aes(y = r_UK, color = "UK 1-month Rate")) +
labs(x = "Date",
     y = "Interest Rate (%)") +
scale_color_manual(values = c("US 1-month Rate" = "blue", "UK 1-month Rate" = "red")) +
theme_minimal()
```

Figure 2: 1-Month Interest Rates: US vs UK



```
# Load required packages
suppressPackageStartupMessages(library(quantmod))
suppressPackageStartupMessages(library(MASS))
suppressPackageStartupMessages(library(Matrix))
suppressPackageStartupMessages(library(numDeriv))

# Extract interest rate data
r_US <- data$r_US # US 1-month rate
r_UK <- data$r_UK # UK 1-month rate

# Define log-likelihood for the CIR model
loglik_CIR <- function(theta, r) {
  mu <- theta[1]
  phi <- theta[2]
  sigma2 <- theta[3]

  T <- length(r)
  r_t <- r[2:T]
  r_t1 <- r[1:(T-1)]

  # Avoid numerical issues
  sigma2 <- max(sigma2, 1e-8)
  phi <- max(min(phi, 0.999), -0.999)
```

```

# Calculate log-likelihood
loglik <- -0.5 * log(2 * pi * r_t1 * sigma2) -
  (r_t - mu * (1 - phi) - phi * r_t1)^2 / (2 * r_t1 * sigma2)

# Return negative log-likelihood (for minimization)
return(-sum(loglik, na.rm = TRUE))
}

# Define log-likelihood for the Vasicek model
loglik_Vasicek <- function(theta, r) {
  mu <- theta[1]
  phi <- theta[2]
  sigma2 <- theta[3]

  T <- length(r)
  r_t <- r[2:T]
  r_t1 <- r[1:(T-1)]

  # Avoid numerical issues
  sigma2 <- max(sigma2, 1e-8)
  phi <- max(min(phi, 0.999), -0.999)

  # Calculate log-likelihood
  loglik <- -0.5 * log(2 * pi * sigma2) -
    (r_t - mu * (1 - phi) - phi * r_t1)^2 / (2 * sigma2)

  # Return negative log-likelihood
  return(-sum(loglik, na.rm = TRUE))
}

# Compute numerical Hessian
compute_hessian <- function(theta, loglik_func, r) {
  hess <- numDeriv::hessian(loglik_func, theta, r = r)
  return(hess)
}

# Estimation function
estimate_model <- function(r, model_name = "CIR") {
  # Initial parameter values
  init_params <- c(mean(r), 0.8, var(r))

  # Select model
  if (model_name == "CIR") {
    loglik_func <- loglik_CIR
    # Parameter constraints: mu > 0, 0 < phi < 1, sigma2 > 0
    lower_bounds <- c(1e-4, 1e-4, 1e-8)
    upper_bounds <- c(Inf, 0.999, Inf)
  } else {
    loglik_func <- loglik_Vasicek
    lower_bounds <- c(-Inf, -0.999, 1e-8)
    upper_bounds <- c(Inf, 0.999, Inf)
  }

  # Maximum likelihood estimation
  result <- optim(
    par = init_params,
    fn = loglik_func,
    r = r,
    method = "L-BFGS-B",

```

```

    lower = lower_bounds,
    upper = upper_bounds,
    hessian = TRUE
)

# Extract estimation results
theta_hat <- result$par
loglik_value <- -result$value
hessian <- result$hessian

# Calculate asymptotic covariance matrix
if (rcond(hessian) > 1e-12) {
  asymp_cov <- solve(hessian)
  # Ensure covariance matrix is symmetric positive definite
  asymp_cov <- nearPD(asymp_cov)$mat
} else {
  warning("Hessian matrix is near singular, using generalized inverse")
  asymp_cov <- ginv(hessian)
}

# Standard errors
std_errors <- sqrt(diag(asymp_cov))

# t-statistics
t_stats <- theta_hat / std_errors

# Results summary
results <- list(
  parameters = theta_hat,
  std_errors = std_errors,
  t_stats = t_stats,
  loglik = loglik_value,
  asymp_cov = asymp_cov,
  hessian = hessian,
  convergence = result$convergence
)

return(results)
}

# Model comparison function
compare_models <- function(cir_results, vasicek_results) {
  comparison <- data.frame(
    Model = c("CIR", "Vasicek"),
    LogLik = c(cir_results$loglik, vasicek_results$loglik),
    AIC = c(
      -2 * cir_results$loglik + 2 * 3,
      -2 * vasicek_results$loglik + 2 * 3
    ),
    BIC = c(
      -2 * cir_results$loglik + 3 * log(length(r_US) - 1),
      -2 * vasicek_results$loglik + 3 * log(length(r_US) - 1)
    )
  )

  return(comparison)
}

```

Table 7: US Interest Rate Model Estimation Results

```
# Estimate models for US interest rates

cat("=== American Interest Rates ===\n")

cir_us <- estimate_model(r_US, "CIR")

vasicek_us <- estimate_model(r_US, "Vasicek")

# Output results

print_results <- function(results, country, model) {

  cat(sprintf("\n%s %s model estimation results:\n", country, model))

  cat("Parameter estimates:\n")

  estimates <- data.frame(

    Parameter = c("mu", "phi", "sigma2"),

    Estimate = results$parameters,

    StdError = results$std_errors,

    tStat = results$t_stats

  )

  print(estimates)

  cat(sprintf("Log-likelihood: %.4f\n", results$loglik))

  cat(sprintf("Convergence status: %d\n", results$convergence))

}

print_results(cir_us, "US", "CIR")

print_results(vasicek_us, "US", "Vasicek")
```

=== American Interest Rates ===

US CIR model estimation results:

Parameter estimates:

	Parameter	Estimate	StdError	tStat
1	mu	0.00010000	10.898336613	9.175712e-06
2	phi	0.99730042	0.002530594	3.940974e+02
3	sigma2	0.04373302	0.001089834	4.012816e+01

Log-likelihood: -236.9732

Convergence status: 0

US Vasicek model estimation results:

Parameter estimates:

	Parameter	Estimate	StdError	tStat
1	mu	5.8889468	2.92671969	2.012132
2	phi	0.9847641	0.01110412	88.684566
3	sigma2	0.5601000	0.04477204	12.510038

Log-likelihood: -353.4131

Convergence status: 0

Table 8: UK Interest Rate Model Estimation Results

```
cat("=== UK Interest Rates ===\n")

cir_uk <- estimate_model(r_UK, "CIR")

vasicek_uk <- estimate_model(r_UK, "Vasicek")

print_results(cir_uk, "UK", "CIR")

print_results(vasicek_uk, "UK", "Vasicek")
```

=== UK Interest Rates ===

UK CIR model estimation results:

Parameter estimates:

	Parameter	Estimate	StdError	tStat
1	mu	8.99627775	5.573881e-06	1.614006e+06
2	phi	0.99721507	6.987061e-03	1.427231e+02
3	sigma2	0.02468564	1.963883e-03	1.256981e+01

Log-likelihood: -198.1268

Convergence status: 0

UK Vasicek model estimation results:

Parameter estimates:

	Parameter	Estimate	StdError	tStat
1	mu	9.1633582	4.225711012	2.168477
2	phi	0.9929275	0.007994119	124.207245
3	sigma2	0.2585271	0.020669879	12.507433

Log-likelihood: -232.4193

Convergence status: 0

Table 9: Model Comparison Results

```
# Model comparison

cat("\n=== Model Comparison ===\n")

comparison_us <- compare_models(cir_us, vasicek_us)

comparison_uk <- compare_models(cir_uk, vasicek_uk)


cat("US Interest Rate Model Comparison:\n")

print(comparison_us)


cat("\nUK Interest Rate Model Comparison:\n")

print(comparison_uk)
```

```
=== Model Comparison ===
US Interest Rate Model Comparison:
  Model   LogLik    AIC    BIC
1    CIR -236.9732 479.9464 491.1850
2 Vasicek -353.4131 712.8262 724.0648

UK Interest Rate Model Comparison:
  Model   LogLik    AIC    BIC
1    CIR -198.1268 402.2537 413.4923
2 Vasicek -232.4193 470.8387 482.0773
```


Table 10: Asymptotic Covariance Matrices

```
# Output asymptotic covariance matrices

cat("\n=== Asymptotic covariance matrices ===\n")

cat("US CIR Model:\n")

print(cir_us$asyp_cov)

cat("\nUS Vasicek Model:\n")

print(vasicek_us$asyp_cov)

cat("\nUK CIR Model:\n")

print(cir_uk$asyp_cov)

cat("\nUK Vasicek Model:\n")

print(vasicek_uk$asyp_cov)
```

```
=== Asymptotic covariance matrices ===
US CIR Model:
3 x 3 Matrix of class "dpoMatrix"
      [,1]      [,2]      [,3]
[1,] 1.187737e+02 -2.533012e-02 8.096863e-04
[2,] -2.533012e-02 6.403906e-06 -1.726767e-07
[3,] 8.096863e-04 -1.726767e-07 1.187737e-06

US Vasicek Model:
3 x 3 Matrix of class "dpoMatrix"
      [,1]      [,2]      [,3]
[1,] 8.565688e+00 -1.027894e-02 -2.998429e-06
[2,] -1.027894e-02 1.233015e-04 4.344832e-09
[3,] -2.998429e-06 4.344832e-09 2.004536e-03

UK CIR Model:
3 x 3 Matrix of class "dpoMatrix"
      [,1]      [,2]      [,3]
[1,] 3.106815e-11 -3.864257e-08 -2.060351e-11
[2,] -3.864257e-08 4.881902e-05 4.451122e-09
[3,] -2.060351e-11 4.451122e-09 3.856837e-06

UK Vasicek Model:
3 x 3 Matrix of class "dpoMatrix"
      [,1]      [,2]      [,3]
[1,] 17.856633554 9.243223e-03 1.893588e-03
[2,] 0.009243223 6.390594e-05 9.795309e-07
[3,] 0.001893588 9.795309e-07 4.272439e-04
```

```

# Prediction function - Generate one-step-ahead predictions
generate_predictions <- function(model_results, r, model_type) {
  mu <- model_results$parameters[1]
  phi <- model_results$parameters[2]

  T <- length(r)
  r_t1 <- r[1:(T-1)]

  if (model_type == "CIR") {
    # CIR model prediction: conditional expectation
    predictions <- mu * (1 - phi) + phi * r_t1
  } else {
    # Vasicek model prediction: conditional expectation
    predictions <- mu * (1 - phi) + phi * r_t1
  }

  return(predictions)
}

# Generate in-sample predictions
cir_pred_us <- generate_predictions(cir_us, r_US, "CIR")
vasicek_pred_us <- generate_predictions(vasicek_us, r_US, "Vasicek")

cir_pred_uk <- generate_predictions(cir_uk, r_UK, "CIR")
vasicek_pred_uk <- generate_predictions(vasicek_uk, r_UK, "Vasicek")

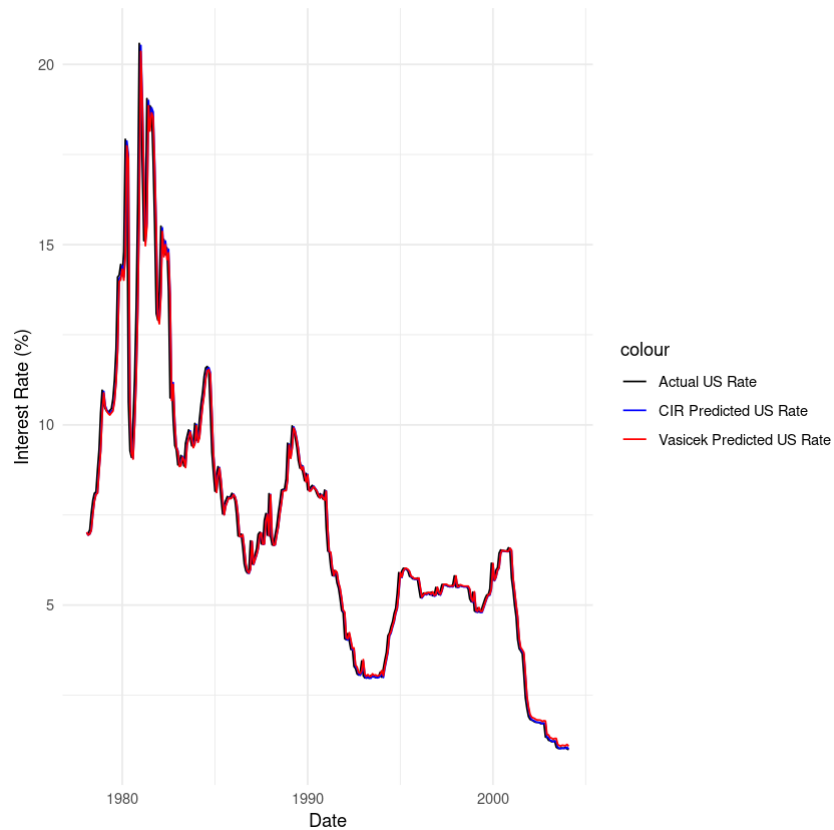
# Create prediction results data frame
actual_us <- r_US[2:length(r_US)]
actual_uk <- r_UK[2:length(r_UK)]

prediction_results <- data.frame(
  Date = data$date[2:nrow(data)],
  Actual_US = actual_us,
  CIR_Pred_US = cir_pred_us,
  Vasicek_Pred_US = vasicek_pred_us,
  Actual_UK = actual_uk,
  CIR_Pred_UK = cir_pred_uk,
  Vasicek_Pred_UK = vasicek_pred_uk
)

# plot predictions vs actuals for US
ggplot(prediction_results, aes(x = Date)) +
  geom_line(aes(y = Actual_US, color = "Actual US Rate")) +
  geom_line(aes(y = CIR_Pred_US, color = "CIR Predicted US Rate")) +
  geom_line(aes(y = Vasicek_Pred_US, color = "Vasicek Predicted US Rate")) +
  labs(x = "Date",
       y = "Interest Rate (%)") +
  scale_color_manual(values = c("Actual US Rate" = "black",
                                "CIR Predicted US Rate" = "blue",
                                "Vasicek Predicted US Rate" = "red")) +
  theme_minimal()

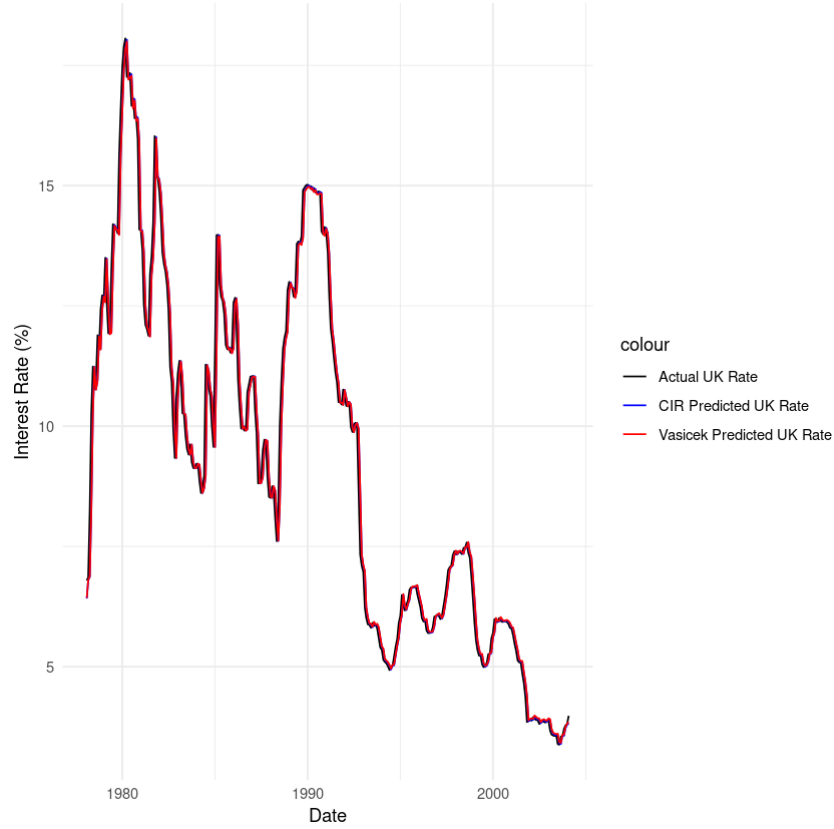
```

Figure 3: US 1-Month Interest Rate: Actual vs Predicted



```
# plot prediction results for UK
ggplot(prediction_results, aes(x = Date)) +
  geom_line(aes(y = Actual_UK, color = "Actual UK Rate")) +
  geom_line(aes(y = CIR_Pred_UK, color = "CIR Predicted UK Rate")) +
  geom_line(aes(y = Vasicek_Pred_UK, color = "Vasicek Predicted UK Rate")) +
  labs(x = "Date",
        y = "Interest Rate (%)") +
  scale_color_manual(values = c("Actual UK Rate" = "black",
                                "CIR Predicted UK Rate" = "blue",
                                "Vasicek Predicted UK Rate" = "red")) +
  theme_minimal()
```

Figure 4: UK 1-Month Interest Rate: Actual vs Predicted



3.1.1 Analysis of CIR and Vasicek Models

Based on the estimation results, the **CIR model** demonstrates superior statistical fit compared to the Vasicek model in both the US and UK interest rate data, with **higher log-likelihood values** (US: -236.97 vs -353.41; UK: -198.13 vs -232.42) and **lower AIC and BIC values**, indicating that the CIR model can better capture the dynamics of interest rates, especially given its characteristic that volatility is related to the level of interest rates, which is more in line with the reality that interest rate fluctuations tend to be amplified at higher levels in the real financial market.

However, the parameter estimation of the US CIR model shows **instability**, such as the long-term mean being close to zero and having a very large standard error, which may reflect the significant fluctuations in US interest rates during the sample period since 1978 (such as during the oil crisis and periods of inflationary pressure), making model identification difficult.

In contrast, the parameter estimation of the UK CIR model is **more stable**, partly due to the relatively mild volatility of UK interest rates during this period, influenced by its stricter financial regulation and gradual monetary policy. Although the Vasicek model has more stable parameter estimation, its assumption of constant volatility limits its goodness of fit, especially in the highly volatile US environment.